



Skill Forge Learning Management System

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Software Requirements Specification (SRS)

Skill Forge Learning Management System

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the Skill Forge Learning Management System (LMS). The platform allows instructors to create courses, students to enroll and learn online, and administrators to manage the platform.

1.2 Scope

The Skill Forge system will:

- Allow students to:
 - Register and Login
 - Enroll in courses
 - Watch lessons and Complete quizzes
- Allow instructors to:
 - create and manage courses
- Allow administrators to:

- Monitor and Manage platform operations
-

2. Overall System Description

2.1 System Architecture (High-Level)

- Client Layer (Web Application – React.js)
- Application Layer (REST API – Node.js + Express)
- Database Layer (MongoDB)

2.2 User Roles

Role	Description
Student	Enrolls and learns courses
Instructor	Creates and manages courses
Admin	Manage users and platform operations

3. Functional Requirements

3.1 User Module

3.1.1 User Registration & Authentication

FR-U1: User Registration

- User shall register with:
 - Full Name

- Email
 - Password
- System shall validate inputs.
- Password must be hashed using bcrypt.

FR-U2: User Login

- User shall login using email and password.
- System shall generate authentication token (JWT).

3.2 Course Management

FR-C1: Create Course

- Instructor shall create course with title, description and price.
- System shall store course information.

FR-C2: Add Lessons

- Instructor uploads video lessons and resources.
- System stores lesson content.

3.3 Enrollment System

FR-E1: Course Enrollment

- Student selects course and enrolls.
- System creates enrollment record.

3.4 Quiz System

FR-Q1: Quiz Creation

- Instructor creates quizzes with MCQ questions.

FR-Q2: Quiz Submission

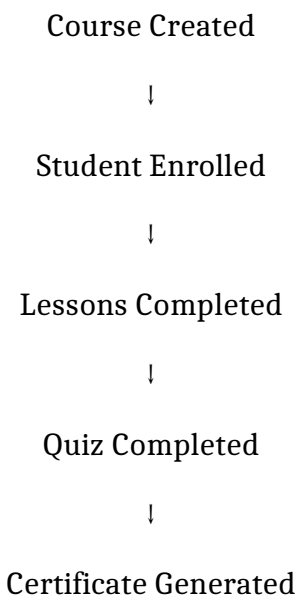
- System automatically grades answers and stores score.

3.5 Certification

FR-CE1: Certificate Generation

- System generates PDF certificate after course completion.
-

4. Learning Status Lifecycle



5. Database Requirements (High-Level Schema)

5.1 Users Collection

Field	Type
id	PK
name	VARCHAR
email	VARCHAR
password	TEXT

role	ENUM
------	------

5.2 Courses Collection

Field	Type
id	PK
title	VARCHAR
description	TEXT
price	INT
instructor_id	FK

6. Non-Functional Requirements

6.1 Performance

- System must support 1000+ users.
- API response time < 2 seconds.

6.2 Security

- Password hashing using bcrypt.
- JWT authentication.
- Role-based access control.

6.3 Availability

- Target uptime 99%.

6.4 Scalability

- Cloud deployable architecture.

6.5 Usability

- Responsive user interface.
-

7. Business Rules

1. Only instructors can create courses.
 2. Only students can enroll in courses.
 3. Certificates are issued only after completion.
 4. Admin can remove inappropriate content.
-

8. Future Enhancements (Optional)

- AI course recommendation
 - Learning analytics dashboard
 - Live classes integration
 - Mobile application version
-

9. Acceptance Criteria

System will be considered complete when:

- Users can register and login.
- Instructors can create courses.
- Students can enroll and complete courses.

- Certificates generate successfully.

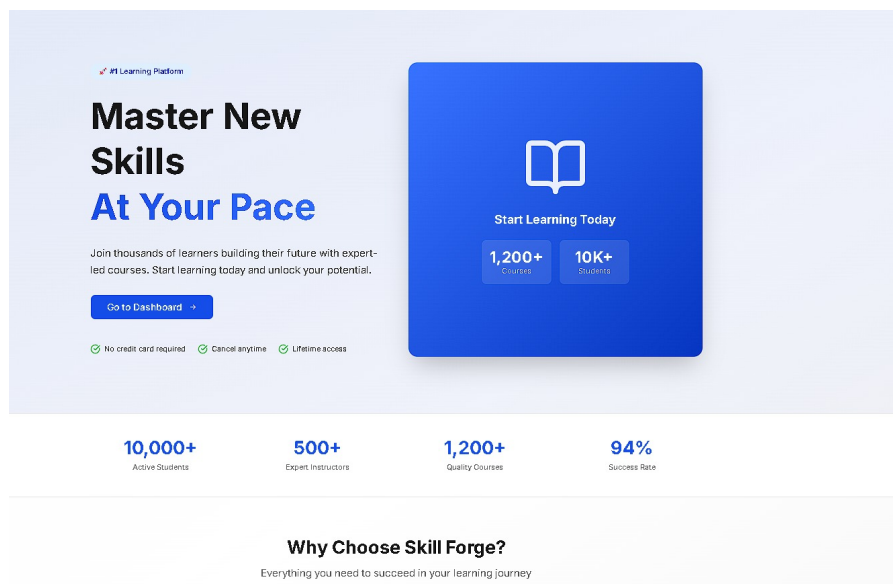
10. Suggested Tech Stack

Layer	Technology
Frontend	React.js
Backend	Node.js / Express
Database	MongoDB
Auth	JWT
Hosting	Vercel / Render

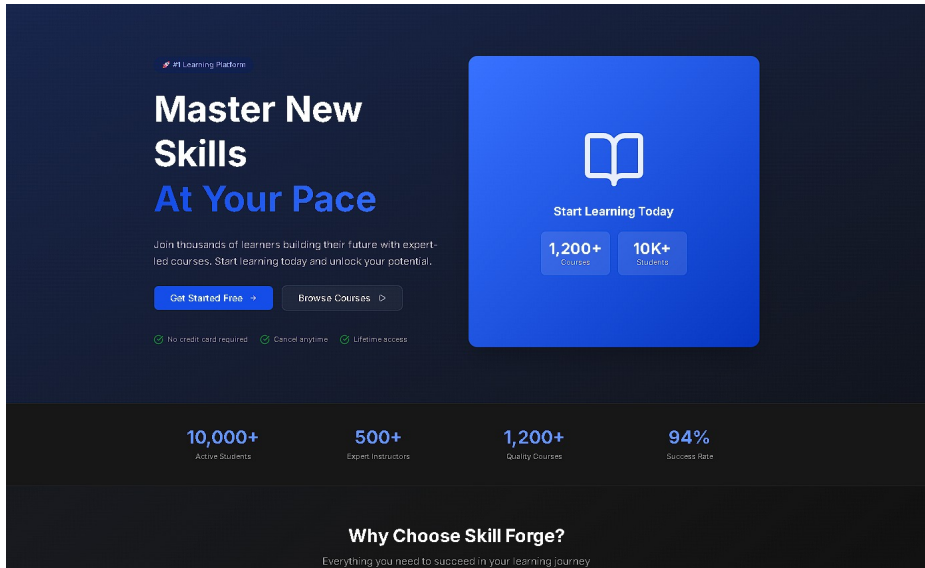
11. Frontend

Main Page

Light Theme



Dark Theme



Sign up page

The sign-up page has a light blue background. At the top center is a blue square icon with a white open book. Below it is the heading "Create your account" and the subtext "Start your learning journey today". The form is a white card with rounded corners. It contains the following fields and elements: "Full Name" with a text input field containing "John Doe"; "Email Address" with a text input field containing "you@example.com"; "I want to" with two buttons: "Learn Student" (highlighted with a blue border) and "Teach Instructor"; "Password" with a text input field containing "*****" and an eye icon; "Confirm Password" with a text input field containing "*****" and an eye icon; a checkbox labeled "I agree to the Terms and Conditions and Privacy Policy"; a blue "Create Account" button; and a link "Already have an account? Sign in". At the bottom center is a link "← Back to Home".

Sign in Page



Create your account

Start your learning journey today

Full Name

 John Doe

Email Address

 you@example.com

I want to



Learn
Student



Teach
Instructor

Password





Confirm Password





☐ I agree to the [Terms and Conditions](#) and [Privacy Policy](#)

Create Account

Already have an account? [Sign in](#)

[← Back to Home](#)

Student Page

Home Page

Skill Forge

[Home](#) [My Courses](#) [Catalog](#) [Certificates](#)

John Doe

Welcome back, John! 🙌

Continue your learning journey and achieve your goals

Total Courses

3

In Progress

2

Completed

1

Learning Hours

88

Continue Learning

Complete Web Development Bootcamp 2024

🕒 45% Complete

▶ Resume Learning →

Upcoming Quizzes

React Masterclass

📅 2026-04-10

pending

Node.js Advanced

📅 2026-04-12

pending

My Courses

[Browse All Courses →](#)

Complete Web Development Bootcamp 2024

Progress

45%

45%

▶ Continue Learning

Modern JavaScript ES6+

Completed

Progress

100%

100%

Certificate

▶ Review

Announcements

New Course: TypeScript Essentials

2 hours ago

Platform maintenance scheduled

1 day ago

Congratulations on completing 5 courses!

3 days ago

Certificates

1

Certificates Earned

Explore Courses

Explore Courses

Discover 0 courses to boost your skills

Category

All Categories

Web Development

Data Science

Mobile Development

Design

Business

Marketing

Photography

Price

All Prices

Free

Paid

Showing 0 courses



No courses found

Try adjusting your search or filters to find what you're looking for.

Clear Filters

Certificate Page



Certificate Not Available

Failed to load certificate. Please try again.

[← Back to Dashboard](#)

Instructor Dashboard

 Skill Forge

[Dashboard](#) [My Courses](#) [Create Course](#)

  Dr. Emily Rodriguez

Instructor Dashboard

Manage your courses and track your teaching performance

Create New Course

Error Loading Data
Using demo instructor data

Total Courses
3

Total Students
8

Total Revenue
\$619.92

Active Courses
3

My Courses Page

My Courses



Published

Complete Web Development Bootcamp 2024

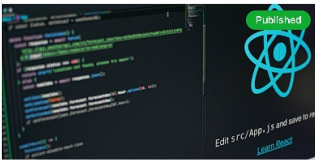
Master web development from scratch. Learn HTML, CSS, JavaScript, React, Node.js, and...

👤 4 \$ \$69.99

Edit Course

▶ Lessons

📄 Quiz



Published

Advanced React & Redux Masterclass

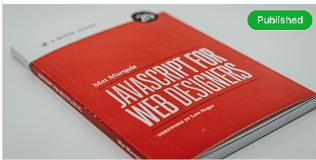
Deep dive into React ecosystem. Learn advanced patterns, hooks, context, Redux, Next.js, and...

👤 2 \$ \$79.99

Edit Course

▶ Lessons

📄 Quiz



Published

Modern JavaScript ES6+

Master modern JavaScript features: arrow functions, destructuring, async/await, modules, a...

👤 2 \$ \$49.99

Edit Course

▶ Lessons

📄 Quiz



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New Course Page

[← Back to Dashboard](#)

Create New Course

Fill in the details below to create your course

Course Title *

e.g., Complete React Masterclass 2024

Description *

Describe what students will learn in this course...

Category *

Select a category

Difficulty Level *

Select difficulty

Price (USD) *

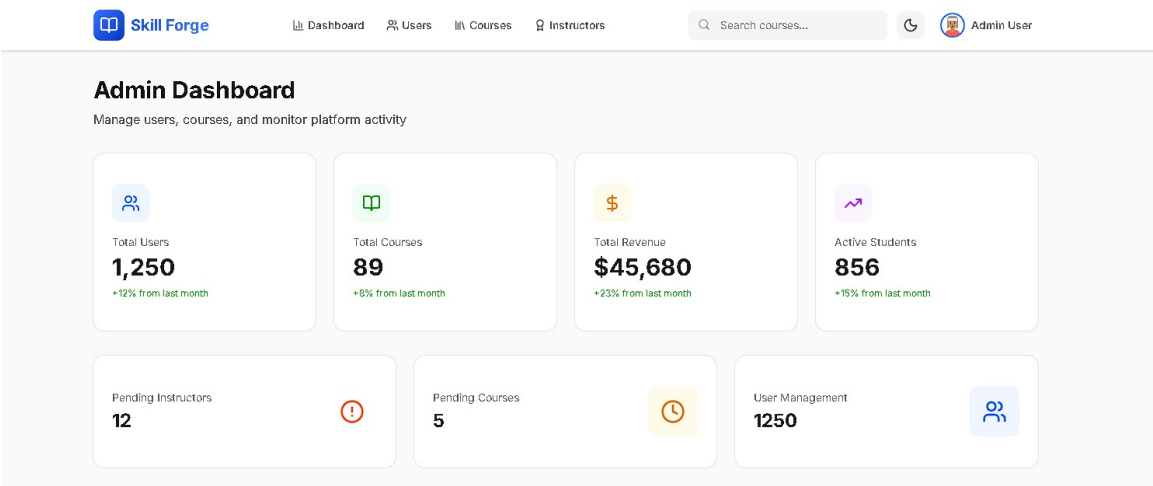
\$ 0.00

Enter 0 for free courses

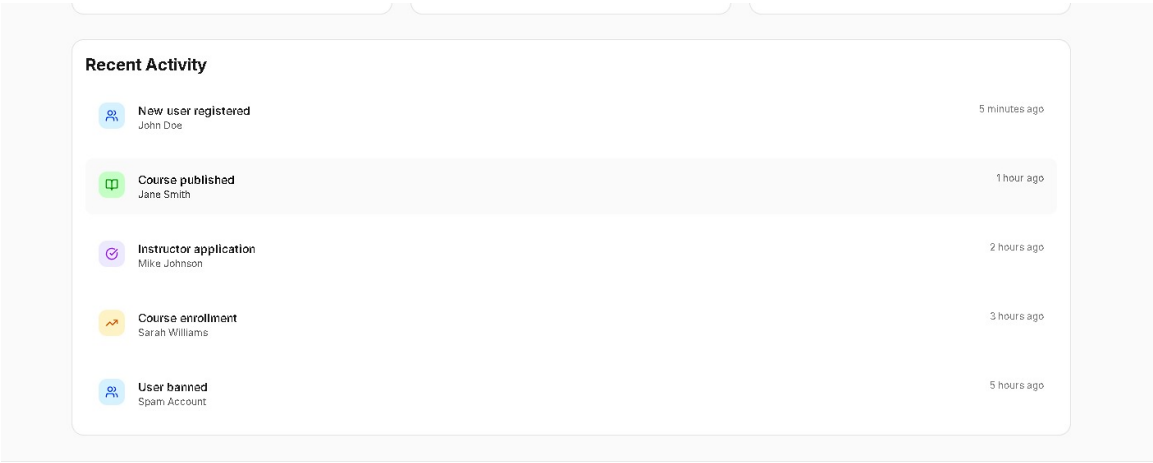
Course Thumbnail

Click to upload course thumbnail

Admin Dashboard



Recent Activities



User Management

Manage user accounts, roles, and permissions

All Roles

All Status

All Users (8)

User	Role	Status	Activity	Joined	Actions
John Doe john@example.com	Student	Active	5 enrollments	15/01/2025	
Jane Smith jane@example.com	Instructor	Active	12 courses	20/11/2025	
Mike Johnson mike@example.com	Student	Active	3 enrollments	10/03/2026	
Sarah Williams sarah@example.com	Instructor	Active	8 courses	05/12/2025	
Spam Account spam@example.com	Student	Banned	0 enrollments	01/04/2026	
Bob Brown bob@example.com	Student	Active	7 enrollments	16/02/2026	

Course Moderation

Monitor and moderate all courses on the platform



Total Courses

5



Total Enrollments

725



Avg Price

\$75



Flagged

1

All Categories



Complete React Masterclass 2026

by Jane Smith

Learn React from scratch with hooks, context, and advanced patterns

Intermediate
Web Development

24S Students
120 Lessons
4.8 Rating

\$89.99

Created 15/01/2026

Details



Python for Data Science

by Mike Johnson

Master Python programming for data analysis and visualization

Beginner
Data Science

189 Students
85 Lessons
4.7 Rating

\$79.99

Created 10/02/2026

Details



Machine Learning A-Z

by Emily Davis

Comprehensive ML course covering all major algorithms

Advanced
Machine Learning

156 Students
150 Lessons
4.9 Rating

\$99.99

Created 05/12/2025

Details

Instructor Control Page

Mobile App Development with React Native

Active

Mobile App Development with React Native

by Chris Anderson

Build iOS and Android apps with React Native

Intermediate Mobile Development

102 Students 95 Lessons 4.6 Rating

\$94.99

Created 01/03/2028

Details

Low Quality Course

Flagged

Low Quality Course

by Spam Instructor

This course has quality issues

Beginner Other

3 Students 5 Lessons 2.1 Rating

\$9.99

Created 04/04/2028

Details

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Made with using +

Pending Application

Skill Forge

Dashboard Users Courses Instructors

Search courses...

Admin User

Instructor Applications

Review and approve instructor applications

Pending (3)

Approved (0)

Rejected (0)

All (3)

Mike Johnson

Pending

mike@example.com

Experienced web developer with 10+ years in the industry. Passionate about teaching React and Node.js.

Expertise: React Node.js MongoDB JavaScript

03/04/2028 Experience: 10 years

View Details

Emily Davis

Pending

emily@example.com

Data scientist and machine learning engineer. Love teaching AI concepts in simple terms.

Expertise: Python Machine Learning TensorFlow Data Science

02/04/2028 Experience: 7 years

View Details

Chris Anderson

Pending

chris@example.com

Mobile app developer specializing in React Native and Flutter. Built 50+ apps.

Expertise: React Native Flutter iOS Android

01/04/2028 Experience: 8 years

View Details

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Approved Applications

Instructor Applications

Review and approve instructor applications

Pending (3)

Approved (1)

Rejected (1)

All (5)

Lisa Wong

lisas@example.com

UX/UI designer with a focus on accessibility and user-centered design.

Expertise:

FigmaUX DesignUIDesignAccessibility

20/03/2028

Experience: 8 years

View Details

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Rejected Applications

Instructor Applications

Review and approve instructor applications

Pending (3)

Approved (1)

Rejected (1)

All (5)

Tom Baker

tom@example.com

Basic knowledge of HTML.

Expertise:

HTML

25/03/2028

Experience: 6 months

View Details

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